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1. Purpose and Scope

This document describes the HL7-protocol used in the BM850-instrument, and how the instrument is supposed to communicate with a Laboratory Information System using HL7.

2. References

[BASE64]	https://tools.ietf.org/html/rfc4648
[HL7]	http://www.hl7.org
[JSON]	http://json.org

3. Changes from previous edition

- Updated 7.5.1 with VET parameters
- Updated 7.5.1 table with Market column
- Updated 7.5.2 with new parameter flags
- Updated example with new data

4. Terms and Definitions

ACK	Acknowledgment
ER7	Encoding system defined in HL7
HL7	Health Level 7
JSON	JavaScript Object Notation
LIS	Laboratory Information System
MLLP	Minimum Lower Layer Protocol, a protocol defined in HL7
TCP/IP	Transmission Control Protocol/Internet Protocol

5. Introduction

The need for communication between a central information system and different medical devices is easily recognized. However, since there are a multitude of manufacturers of both medical devices and of information systems, there's a need of a standardized protocol used for the communication. HL7, [HL7], is standard for electronic data exchange in healthcare environments, which is becoming used more and more and customers are starting to ask for this protocol to be implemented in medical devices.

6. HL7 in BM850

Only sample results can be transmitted via HL7, and data is only sent in one direction (disregarding ACK:s), from the instrument to a LIS.

6.1. Version

The version of HL7 used by the BM850 instrument is 2.7.

6.2. Protocol

MLLP over TCP/IP is used as protocol.

6.3. Data-format

The data-format used is ER7, also known as "pipe-delimited".

7. Sample results

A sample message is composed of several different segments. An example message is shown in Appendix A.

Segment type	Segment name	Repetitions	Description
MSH	Message Header	1	Identifies the meaning of the message.
SFT	Software Segment	1	Information about the SW version in the instrument.
OBR	Observation Request	1	Identifies the sample.
NTE	Notes and Comments	1	User comment for the sample.
OBX	Observation Result	1-n	One segment for each item in the sample results

7.1. Message Header, MSH

Field	Name	Description

1	Field separator	`'|'`
2	Encoding Characters	`'^~/&'`
3	Sending Application	User configurable. Default is 'BM850^HL7MW'. Max 40 chars.
4-6		Empty
7	Date/Time of Message	Date/Time when the message was sent according to the instrument. 14 chars, YYYYMMDDHHMMSS.
8		Empty
9	Message Type	`'ORU^R01'`
10	Message Control ID	Set by the instrument to identify ACK. Max 20 chars.
11	Processing ID	`'P'`
12	Version ID	`'2.7'`
13-17		Empty
18	Character Set	`'UNICODE UTF-8'`

7.2. Software Segment, SFT

Field	Name	Description
1	Software Vendor Organization	`'Boule Medical AB'`
2	Software Certified Version or Release Number	Varies depending on the SW version. Max 50 chars.
3	Software Product Name	Varies depending on instrument model. Max 50 chars.
4	Software Binary ID	Repeat of field 2.

7.3. Observation Request, OBR

Field	Name	Description
1	Set ID	`'1'`
2		Empty
3	Filler Order Number	The sequence number of the sample as defined by the instrument. 1-99999.
4	Universal Service Identifier	Sample ID 1. Max 50 chars.
5-21		Empty
22	Result Report/Status Change Date/Time	Date and time when the sample was analyzed. 12 chars, YYYYMMDDHHMM.

7.4. Notes and Comments, NTE

Field	Name	Description
1	Set ID	`'1'`
2		Empty
3	Comment	User comment for the sample. Max 100 chars.

7.5. Observation Result, OBX

Field	Name	Description
1	Set ID	1-*n*, where *n* is the number of OBX-segments in the message.
2	Value Type	See chapter 7.5.1
3	Observation Identifier	See chapter 7.5.1
4		Empty
5	Observation Value	See chapter 7.5.1
6	Units	See chapter 7.5.1
7	Reference Range	See chapter 7.5.1
8	Interpretation Codes	See chapter 7.5.2
9-10		Empty
11	Observation Result Status	`'P'`

7.5.1. Data items in a sample results

The following table describes each item that can be sent in an OBX-segment. Only relevant items are sent, so if for instance the parameter *RBC* is not present in the sample results no OBX with Observation Identifier 'RBC' will be sent.

Note that the parameter names used in the table are the standard ones, and not the user-configured ones.

BM850	Observation Identifier	Observation Identifier	Value Type	Unit	Market (All,	Description

	Result	Alert Limit *			Human or Vet)	
RBC	RBC	RBCAL	NM	10*12/L	All	
MCV	MCV	MCVAL	NM	fL	All	
HCT	HCT	HCTAL	NM	%	All	
MCH	MCH	MCHAL	NM	pg	All	
MCHC	MCHC	MCHCAL	NM	g/dL	All	
RDW	RDWR	RDWRAL	NM	%	All	
RDWa	RDWA	RDWAAL	NM	fL	All	
PLT	PLT	PLTAL	NM	10*9/L	All	
MPV	MPV	MPVAL	NM	fL	All	
PCT	PCT	PCTAL	NM	%	All	
PDW	PDW	PDWAL	NM	fL	All	
PDW%	PDWR	PDWRAL	NM	%	Human	Configurable in BM850 instrument
LPCR/P-LCR	LPCR or P-LCR	LPCRAL or P-LCRAL	NM	%	Human	Configurable in BM850 instrument
LPCA/P-LCC	LPCA or P-LCC	LPCAAL or P-LCCAL	NM	10*9/L	Human	Configurable in BM850 instrument
HGB	HGB	HGBAL	NM	g/dL	All	
WBC	WBC	WBCAL	NM	10*9/L	All	
LYM	LYMA	LYMAAL	NM	10*9/L	All	
MID	MIDA	MIDAAL	NM	10*9/L	Human	
MON	MONA	MONAAL	NM	10*9/L	Vet	
GRA	GRNA	GRNAAL	NM	10*9/L	All	When 3-part diff
NEU	NEUA	NEUAAL	NM	10*9/L	Vet	When 4-part diff
EOS	EOSA	EOSAAL	NM	10*9/L	Vet	When 4-part diff
LYM%	LYMR	LYMRAL	NM	%	All	
MID%	MIDR	MIDRAL	NM	%	Human	
MON%	MONR	MONRAL	NM	%	Vet	
GRA%	GRNR	GRNRAL	NM	%	All	When 3-part diff
NEU%	NEUR	NEURAL	NM	%	Vet	When 4-part diff
EOS%	EOSR	EOSRAL	NM	%	Vet	When 4-part diff
Sample ID 2	ID2	-	ST		All	Max 50 chars
Profile	PROF	-	ST		All	Max 50 chars
Method	METH	-	ST		All	See chapter 7.5.3
Operator	OPID	-	ST		All	Max 10 chars
Histogram	HGRAM	-	ED		All	See chapter 7.5.4

Table 1: Data items in sample OBX

* Alert Limits: Fields 2, 5 and 8 are empty. Only fields 6 and 7 contain any data.

7.5.2. Parameter flags

Possible values in field 8 in a sample OBX-segment. Only applicable for OBX-segments containing parameter values. Note that some of the flags below are only possible for specific markets.

Parameter Flag	Description
OR	Measuring chamber indicator, Measurement warning
DF	Reagent pipette indicator, Diluent system problem
DP	Reagent pipette indicator, Diluent system problem
LF	Reagent pipette indicator, Lyse system problem
LP	Reagent pipette indicator, Lyse system problem
TE	Mixing beaker indicator, Liquid system problem
TL	Reagent pipette indicator, Possible orifice blockage
TU	Reagent pipette indicator, Possible orifice blockage
ST	Reagent pipette indicator, Air bubbles
TB	Reagent pipette indicator, Air bubbles
HT	Temperature Indicator, Instrument temperature outside limits
HL	HGB indicator, HGB measuring problem

HH	HGB indicator, HGB measuring problem
HF	HGB indicator, HGB measuring problem
HO	HGB indicator, HGB measuring problem
HS	HGB indicator, HGB measuring problem
HN	HGB indicator, HGB measuring problem
ML	Out-of-Range Indicator, Parameter below measurement range
SE	Measuring chamber indicator, Measurement statistics warning
DE	Distribution indicator, Small particle interference
FD	Distribution indicator, Irregular distribution PLT/RBC. Re-analyze sample.
NM	WBC differential abnormality, No mode found for WBC
OM	WBC differential abnormality, Only one WBC population found
O2	WBC differential abnormality, Only one WBC population found
TM	WBC differential abnormality, More than two modes found in WBC distribution
BD	WBC differential abnormality, High interference between populations
RP	PLT-C profile indicator, Red cells present
AF	Aspiration indicator (sample probe), Aspiration failed
NR	Reagent and control indicator, Enter new reagent in Reagent setup menu
MB	PLT-C profile indicator, Missing background count
BL	Out-of-Range Indicator, Parameter below linear range
AL	Out-of-Range Indicator, Parameter above linear range
ER	Reagent and control indicator, Reagent is expired
EC	Reagent and control indicator, Control is expired
HW	HGB indicator, HGB measurement warning
EF	EOS reagent filling problem
EP	EOS reagent emptying problem
LW	Low WBC
LM	Lym mode. Only a LYM population found
MM	Mono Mode. Only a MON population found
GM	Gran Mode. Only a GRA population found
L2	Measurement warning. The sample may contain lyse resistant WBC
MH	Out-of-range indicator. Parameter above measurement range
SC	PLT indicator. Suspected PLT concentrate
XF	PLT indicator. Extended counting time failure

Table 2: Parameter flags

7.5.3. Method

Possible values for field 5 in a sample OBX-segment where the Observation Identifier is 'METH'. This is the aspiration method used when analyzing the sample.

Method	Description
OT	Open Tube
PD	Predilute
MC	Micro-capillary
CP	Cap Piercer
AS#xx	Auto Sample. *xx* is the position on the auto-sampler wheel.

Table 3: Aspiration methods

7.5.4. Histograms

Every sample results can contain a number of histograms. These histograms are formatted as a JSON-string [JSON], which is then encoded with Base64, [BASE64]. The encoded string is sent in the *Observation Value* field in the OBX-segment with identifier *HGRAM*.

The format of the JSON-string is as follows:

```
'<name>': {'count': <number_c>, 'min': <number_m>, 'max': <number_x>, 'filter': <number_f>, 'disc': [<number_d>, …], 'values': {'<name_1>': [<number_v>,…], <next_histg>}}
```

Tag	Value type	Description
<name>	String	Name of histogram

<number_c>	Integer	Specifies number of values in the histogram
<number_m>	Integer	Specifies the cell volume of the cells that are represented by the smallest interval in the histogram
<number_x>	Integer	Specifies the cell volume of the cells that are represented by the largest interval in the histogram
<number_f>	Integer	Filter factor used for the data-set
[<number_d>, …]	Array of integers	Specifies the index of the data-set for discriminators
<name_1>	String	Name of the histogram. If there's only one data-set it's the same as <name>
[<number_v>, …]	Array of integers	Specifies the data value for the data-set

Example with three histograms: One for PLT and one for RBC with one data-set each, and one for WBC with 3 data-sets (LYM, MID, GRA).


```
{'PLT': {'count': 80, 'min': 0, 'max': 30, 'filter': 7, 'disc': [66], 'values': {'PLT': [0, 0, 0, 0,
0, 0, 0, 1, 1, 2, 3, 4, 5, 7, 8, 11, 13, 16, 19, 22, 25, 28, 31, 33, 34, 35, 35, 34, 33, 32, 30, 28,
27, 25, 23, 21, 20, 18, 16, 14, 13, 12, 11, 9, 8, 8, 7, 6, 5, 4, 4, 3, 3, 2, 2, 2, 2, 2, 2, 2, 2, 1,
1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1]}}, 'RBC': {'count': 80, 'min': 0, 'max': 250,
'filter': 2, 'disc': [10], 'values': {'RBC': [0, 3, 18, 44, 58, 48, 27, 13, 6, 3, 2, 2, 5, 11, 24,
47, 80, 118, 159, 194, 224, 245, 254, 247, 226, 199, 178, 163, 144, 123, 107, 98, 93, 87, 79, 68,
63, 58, 49, 41, 35, 29, 24, 20, 16, 13, 10, 8, 7, 6, 6, 4, 4, 4, 3, 2, 2, 2, 2, 1, 1, 1, 1, 1, 0, 0,
0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0]}}, 'WBC': {'count': 80, 'min': 0, 'max': 450, 'filter': 4,
'disc': [8], 'values': {'LYM': [0, 0, 0, 1, 4, 10, 18, 28, 37, 43, 46, 44, 40, 34, 28, 23, 19, 16,
15, 13, 11, 9, 6, 4, 2, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0], 'MID':
[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 2, 4, 5, 5, 5, 4, 4, 3, 3, 3, 2, 2,
2, 1, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0], 'GRA': [0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 0, 0, 0, 0, 1, 1, 1, 2, 2, 2, 3, 4, 5, 7, 9, 12, 14, 17, 19, 20, 21, 21, 21, 20, 20, 19,
18, 16, 15, 14, 13, 11, 10, 9, 8, 8, 7, 7, 6, 5, 5, 4, 4, 4, 4, 4, 4, 3, 3, 3, 3, 2, 2, 2, 2, 2, 2,
2, 2, 2, 2]}}}
```


The resulting OBS-segment would then be:


```
OBX|22|ED||HGRAM|eyJQTFQiOiB7ImNvdW50IjogODAsICJtaW4iOiAwLCAibWF4IjogMzAsICJmaWx0ZXIiOiA3LCAiZGlzYyI
6IFs2Nl0sICJ2YWx1ZXMiOiB7IlBMVCI6IFswLCAwLCAwLCAwLCAwLCAwLCAwLCAxLCAxLCAyLCAzLCA0LCA1LCA3LCA4LCAxMSw
gMTMsIDE2LCAxOSwgMjIsIDI1LCAyOCwgMzEsIDMzLCAzNCwgMzUsIDM1LCAzNCwgMzMsIDMyLCAzMCwgMjgsIDI3LCAyNSwgMjM
sIDIxLCAyMCwgMTgsIDE2LCAxNCwgMTMsIDEyLCAxMSwgOSwgOCwgOCwgNywgNiwgNSwgNCwgNCwgMywgMywgMiwgMiwgMiwgMiw
gMiwgMiwgMiwgMiwgMSwgMSwgMSwgMSwgMSwgMSwgMSwgMSwgMSwgMSwgMSwgMSwgMSwgMSwgMSwgMSwgMSwgMSwgMV19fSwgIlJ
CQyI6IHsiY291bnQiOiA4MCwgIm1pbiI6IDAsICJtYXgiOiAyNTAsICJmaWx0ZXIiOiAyLCAiZGlzYyI6IFsxMF0sICJ2YWx1ZXM
iOiB7IlJCQyI6IFswLCAzLCAxOCwgNDQsIDU4LCA0OCwgMjcsIDEzLCA2LCAzLCAyLCAyLCA1LCAxMSwgMjQsIDQ3LCA4MCwgMTE
4LCAxNTksIDE5NCwgMjI0LCAyNDUsIDI1NCwgMjQ3LCAyMjYsIDE5OSwgMTc4LCAxNjMsIDE0NCwgMTIzLCAxMDcsIDk4LCA5Myw
gODcsIDc5LCA2OCwgNjMsIDU4LCA0OSwgNDEsIDM1LCAyOSwgMjQsIDIwLCAxNiwgMTMsIDEwLCA4LCA3LCA2LCA2LCA0LCA0LCA
0LCAzLCAyLCAyLCAyLCAyLCAxLCAxLCAxLCAxLCAxLCAwLCAwLCAwLCAwLCAwLCAwLCAwLCAwLCAwLCAwLCAwLCAwLCAwLCAwLCA
wLCAwXX19LCAiV0JDIjogeyJjb3VudCI6IDgwLCAibWluIjogMCwgIm1heCI6IDQ1MCwgImZpbHRlciI6IDQsICJkaXNjIjogWzh
dLCAidmFsdWVzIjogeyJMWU0iOiBbMCwgMCwgMCwgMSwgNCwgMTAsIDE4LCAyOCwgMzcsIDQzLCA0NiwgNDQsIDQwLCAzNCwgMjg
sIDIzLCAxOSwgMTYsIDE1LCAxMywgMTEsIDksIDYsIDQsIDIsIDEsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDA
sIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDA
sIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDBdLCAiTUlEIjogWzAsIDAsIDAsIDAsIDA
sIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDEsIDIsIDQsIDUsIDUsIDUsIDQsIDQsIDMsIDM
sIDMsIDIsIDIsIDIsIDEsIDEsIDEsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDA
sIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDB
dLCAiR1JBIjogWzAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDAsIDA
sIDAsIDEsIDEsIDEsIDIsIDIsIDIsIDMsIDQsIDUsIDcsIDksIDEyLCAxNCwgMTcsIDE5LCAyMCwgMjEsIDIxLCAyMSwgMjAsIDI
wLCAxOSwgMTgsIDE2LCAxNSwgMTQsIDEzLCAxMSwgMTAsIDksIDgsIDgsIDcsIDcsIDYsIDUsIDUsIDQsIDQsIDQsIDQsIDQsIDQ
sIDMsIDMsIDMsIDMsIDIsIDIsIDIsIDIsIDIsIDIsIDIsIDIsIDIsIDJdfX19||||||P
```


8. ACK

When using HL7, ACK must be enabled and the HL7-mechanism called "Original Acknowledgement Mode" will be used. This means that the LIS shall respond with a message containing one MSH-, one MSA- and if needed one ERR-segment as specified below.

8.1. Message Header, MSH

Field	Name	Description
9	Message Type	'ACK^R01'

The other fields are defined by the LIS.

8.2. Message Acknowledgement, MSA

Field	Name	Description
1	Acknowledgement Code	See [HL7]
2	Message Control ID	Shall correspond to the value of field 10 in the MSH-segment in the message that is acknowledged.

8.3. Error Segment, ERR

Field	Name	Description
1-2		Empty
3	HL7 Error Code	See [HL7]. The minimum is that the code is specified, i.e. `'200'` for "Unsupported Message Type"
4	Severity	See [HL7]

Appendix A: Example HL7-message for sample results

```
<SB>MSH|^~\&|BM850^HL7MW||||20180323135226|ORU^R01|BM_1|P|2.7||||||UNICODE UTF-8<CR>
SFT|Boule Medical AB|r14869 trunk Devel|M|r14869 trunk Devel<CR>
OBR|1||1|Charlie|||||||||||||||||201803231153<CR>
NTE|1||""<CR>
OBX|1|ST|ID2||""||||||P<CR>
OBX|2|ST|PROF||Blood||||||P<CR>
OBX|3|ST|METH||OT||||||P<CR>
OBX|4|ST|OPID||Operator||||||P<CR>
OBX|5|NM|RBC||4.67|10*12/L|3.50-5.50|""|||P<CR>
OBX|6|NM|MCV||89.6|fL|75.0-100.0|""|||P<CR>
OBX|7|NM|HCT||41.9|%|35.0-55.0|""|||P<CR>
OBX|8|NM|MCH||30.2|pg|25.0-35.0|""|||P<CR>
OBX|9|NM|MCHC||33.7|g/dL|31.0-38.0|""|||P<CR>
OBX|10|NM|RDWR||13.9|%|11.0-16.0|""|||P<CR>
OBX|11|NM|RDWA||67.0|fL|0.1-250.0|""|||P<CR>
OBX|12|NM|PLT||216|10*9/L|130-400|""|||P<CR>
OBX|13|NM|MPV||7.9|fL|6.5-11.0|""|||P<CR>
OBX|14|NM|PCT||0.17|%|0.01-9.99|""|||P<CR>
OBX|15|NM|PDW||11.7|fL|0.1-30.0|""|||P<CR>
OBX|16|NM|PDWR||28.9|%|0.1-99.9|""|||P<CR>
OBX|17|NM|P-LCR||16.9|%|0.1-99.9|""|||P<CR>
OBX|18|NM|P-LCC||37|10*9/L|1-1999|""|||P<CR>
OBX|19|NM|HGB||14.1|g/dL|11.5-16.5|""|||P<CR>
OBX|20|NM|WBC||5.3|10*9/L|3.5-10.0|""|||P<CR>
OBX|21|NM|LYMA||1.9|10*9/L|0.9-5.0|""|||P<CR>
OBX|22|NM|MIDA||0.4|10*9/L|0.1-1.5|""|||P<CR>
OBX|23|NM|GRNA||3.0|10*9/L|1.2-8.0|""|||P<CR>
OBX|24|NM|LYMR||35.8|%|15.0-50.0|""|||P<CR>
OBX|25|NM|MIDR||7.4|%|2.0-15.0|""|||P<CR>
OBX|26|NM|GRNR||56.8|%|35.0-80.0|""|||P<CR>
OBX|27|ED||HGRAM|eyJQTFQiOiB7ImNvdW50IjogODAsICJtaW4iOiAwLCAibWF4IjogMzAsICJmaWx0ZXIiOiA3
LCAiZGlzYyI6IFs2M10sICJ2YWx1ZXMiOiB7IlBMVCI6IFswLCAwLCAwLCAwLCAyLCAzLCA2LCA5LCAxMiwgMTYsI
DE5LCAyMSwgMjMsIDIzLCAyMywgMjIsIDIxLCAyMCwgMTgsIDE3LCAxNiwgMTUsIDE0LCAxMywgMTIsIDExLCAxMC
wgMTAsIDksIDgsIDgsIDgsIDcsIDcsIDYsIDYsIDUsIDUsIDQsIDQsIDMsIDMsIDIsIDIsIDIsIDIsIDIsIDIsIDI
sIDIsIDIsIDEsIDEsIDEsIDEsIDEsIDEsIDAsIDAsIDAsIDAsIDAsIDAsIDEsIDEsIDEsIDEsIDEsIDEsIDEsIDEs
IDEsIDEsIDEsIDEsIDEsIDEsIDEsIDEsIDFdfX0sICJSQkMiOiB7ImNvdW50IjogODAsICJtaW4iOiAwLCAibWF4I
jogMjUwLCAiZmlsdGVyIjogMiwgImRpc2MiOiBbOF0sICJ2YWx1ZXMiOiB7IlJCQyI6IFszLCAxOSwgMzUsIDMxLC
AyMCwgMTIsIDcsIDMsIDIsIDEsIDIsIDIsIDIsIDIsIDMsIDYsIDExLCAyNCwgNDYsIDc5LCAxMjMsIDE3MSwgMjEyLCA
yNDIsIDI1NCwgMjQ0LCAyMjYsIDIwNCwgMTc1LCAxNDYsIDEyMywgMTA3LCA5NCwgODUsIDgyLCA4MCwgNzYsIDc0
LCA3NCwgNzMsIDY3LCA2NCwgNTgsIDQ5LCA0NCwgMzksIDMyLCAyNywgMjIsIDE5LCAxNSwgMTIsIDEwLCA4LCA2L
CA1LCA0LCA0LCAzLCAzLCAyLCAyLCAyLCAyLCAyLCAyLCAyLCAxLCAxLCAxLCAxLCAxLCAwLCAwLCAwLCAwLCAwLC
AwLCAwLCAwLCAwXX19LCAiV0JDIjogeyJjb3VudCI6IDgwLCAibWluIjogMCwgIm1heCI6IDQwMCwgImZpbHRlciI
6IDQsICJkaXNjIjogWzhdLCAidmFsdWVzIjogeyJMWU0iOiBbMCwgMCwgMCwgMCwgMCwgMSwgMiwgMywgNCwgNiwg
OSwgMTMsIDE3LCAyMCwgMjIsIDIyLCAyMCwgMTgsIDE2LCAxNSwgMTMsIDExLCA4LCA2LCA0LCAzLCAyLCAxLCAxL
CAwLCAwLCAwLCAwLCAwLCAwLCAwLCAwLCAwLCAwLCAwLCAwLCAwLCAwLCAwLCAwLCAwLCAwLCAwLCAwLCAwLCAwLC
AwLCAwLCAwLCAwLCAwLCAwLCAwLCAwLCAwLCAwLCAwLCAwLCAwLCAwLCAwLCAwLCAwLCAwLCAwLCAwLCAwLCAwLCA
wLCAwLCAwLCAwLCAwLCAwLCAwXSwgIk1JRCI6IFswLCAwLCAwLCAwLCAwLCAwLCAwLCAwLCAwLCAwLCAwLCAwLCAw
LCAwLCAwLCAwLCAwLCAwLCAwLCAwLCAwLCAxLCAxLCAxLCAyLCAyLCAyLCAzLCAzLCAzLCAzLCAzLCAyLCAyLCAyL
CAxLCAxLCAwLCAwLCAwLCAwLCAwLCAwLCAwLCAwLCAwLCAwLCAwLCAwLCAwLCAwLCAwLCAwLCAwLCAwLCAwLCAwLC
AwLCAwLCAwLCAwLCAwLCAwLCAwLCAwLCAwLCAwLCAwLCAwLCAwLCAwLCAwLCAwLCAwLCAwLCAwLCAwLCAwLCAwLCA
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```

```
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9fX0=||||||P<CR>
<EB><CR>
```

Where: <SB>=Start Block=0x0B, <CR>=Carriage Return=0x0D, <EB>=End Block=0x1c